

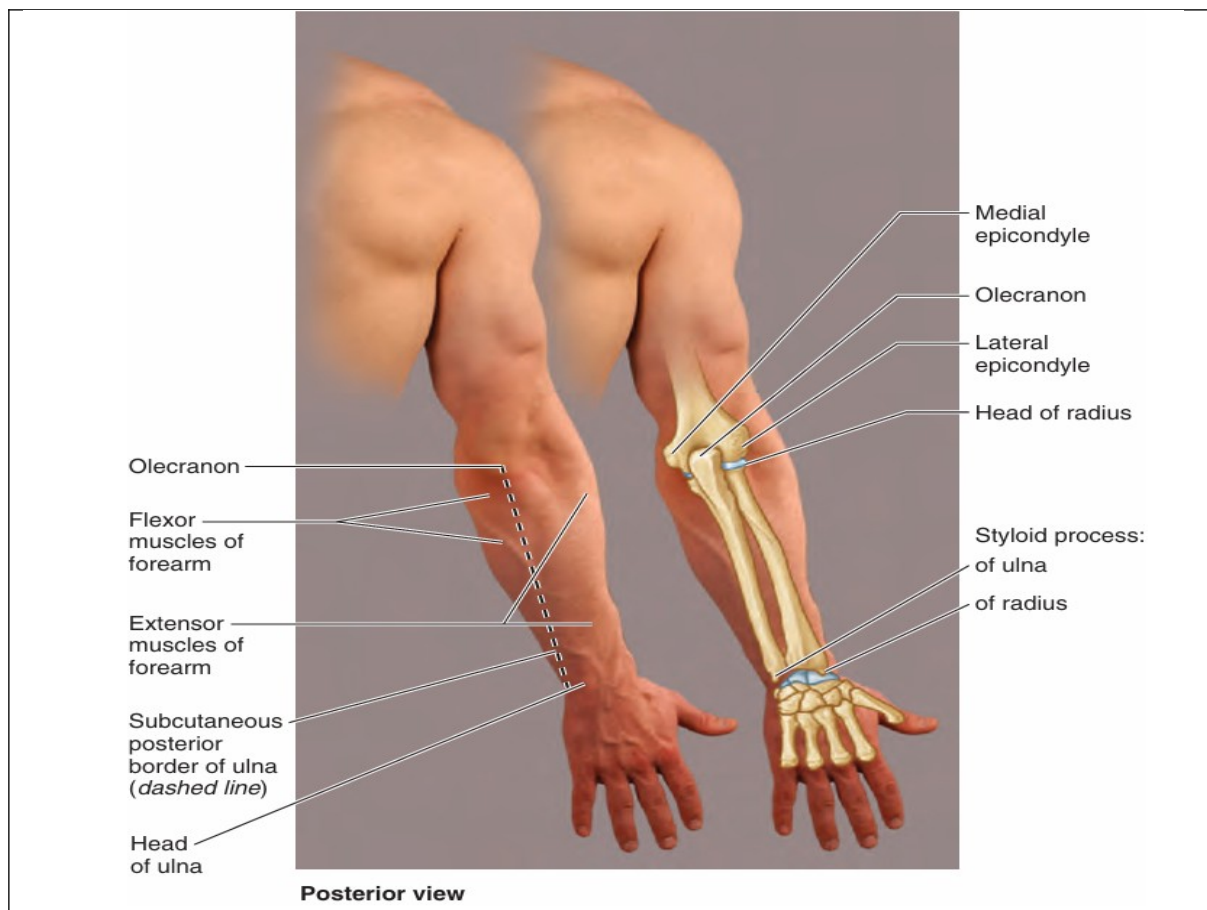
## Skill lab

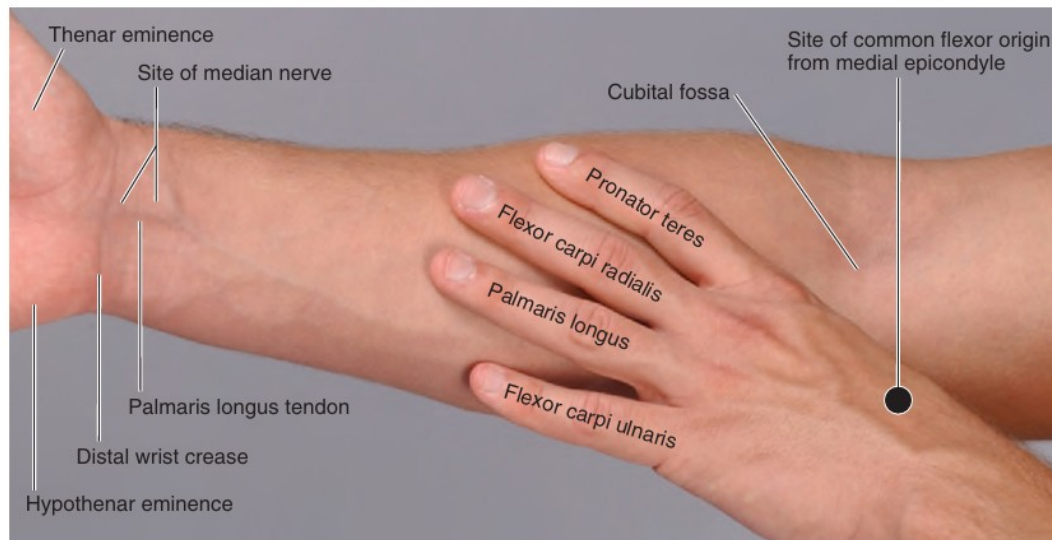
### Unit 3: musculoskeletal system

#### Week 4: Elbow joint and ulnar nerve examination

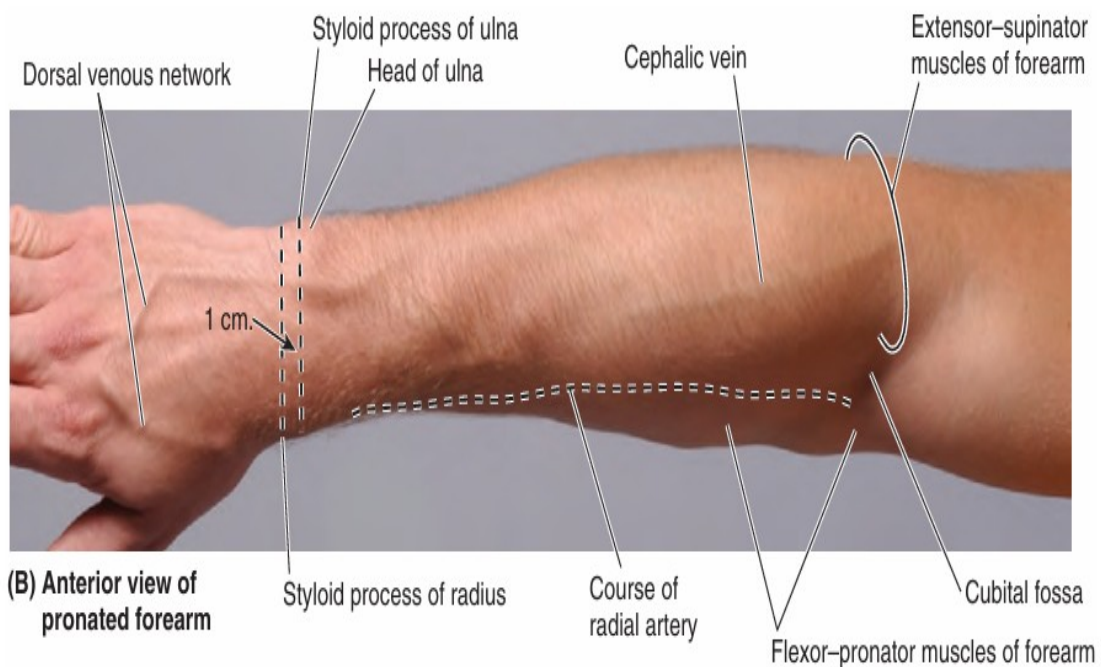
##### Objectives:

- 1- to know the surface anatomy of the elbow joint
- 3- how to take a focused history from patient with elbow pain
- 4- how to perform physical examination of the elbow joint
- 5- how to perform ulnar nerve examination
- 6- how to interpret findings from history and physical examination from the patient (clinical reasoning)





(A) Anterior view of supinated forearm



(B) Anterior view of pronated forearm

**FIGURE 6.72.** (Continued) Surface anatomy of anterior forearm.

### The physical examination:

The elbow joint has humeroulnar, radiocapitellar and superior radioulnar articulations. The medial and lateral epicondyles are the common flexor and extensor origins, respectively, for the forearm muscles. These two prominences and the tip of the olecranon should be easily palpated. They form an equilateral triangle when the elbow is flexed to 90 degrees, and a straight line when the elbow is fully extended.

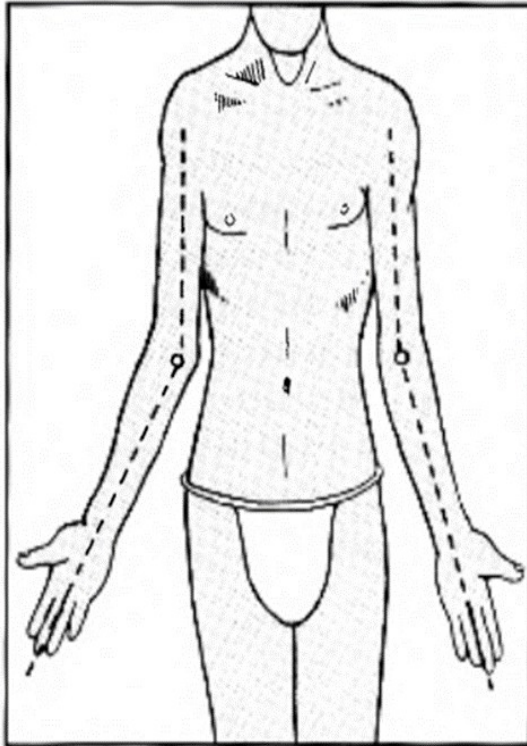
### Examination sequence

- Look at the overall alignment of the extended elbow. There is normally a valgus angle of 11–13 degrees with the elbow fully extended (the 'carrying angle').

### 1- Look

### 2- Feel: look and feel for

- The swelling of synovitis between the lateral epicondyle and olecranon, resulting in a block to full extension
- Skin changes of psoriasis
- Rheumatoid nodules on the proximal extensor surface of the forearm.
- Palpate the bony contours of the lateral and medial epicondyles and olecranon tip.
- Feel for olecranon bursa swelling, nodules and tophi.



**5.4. Inspection (4):** Ask the patient to extend both elbows and note the carrying angle on both sides. Any small difference between the sides will then be obvious.



**FIGURE 1.44**

Large rheumatoid nodules are seen in a classic location along the extensor surface of the forearm and in the olecranon bursa

### 3- Move

- Assess the extension–flexion arc: ask the patient to touch their shoulder on the same side and then straighten the elbow as far as possible.

- Assess supination and pronation: ask the patient to put their elbows by the sides of their body and flex them to 90 degrees. Now ask them to turn their palms upwards (supination) and then downwards (pronation).

| Nerve  | Motor function                                                                           | Sensory function                                                                                     |
|--------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| Median | Wrist flexion and abduction<br>Thumb and index finger flexion<br>Thumb opposition        | Palmar thumb, index and middle finger, and radial half of ring finger                                |
| Ulnar  | Wrist flexion and adduction<br>Finger spread<br>Flexion of distal phalanx of fifth digit | Ulnar side of ring finger and associated dorsal hand<br><br>Little finger and associated dorsal hand |
| Radial | Wrist extension<br>Supination<br>Thumb extension                                         | Dorsal web space between thumb and index finger and dorsal hand, excluding the ulnar distribution    |

The following tests establish motor function of the major nerves with minimal extremity movements:

- Anterior interosseous nerve (branch of median nerve) and radial nerve – "OK" sign
- Ulnar nerve – Finger spread against resistance or holding a piece of paper firmly between the middle and ring fingers
- Radial nerve – Thumb's up sign

The clinician should test the following sites to evaluate peripheral nerve sensation:

- Dorsal web space between thumb and forefinger – Radial nerve
- Palmar index finger – Median nerve
- Little finger – Ulnar nerve

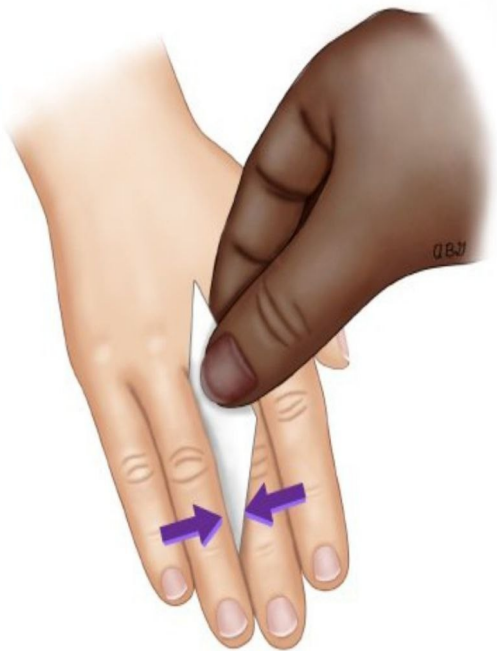
**For patients with suspected displaced supracondylar fracture of humerus, do not perform movement of the elbow joint without stabilization.**

## **OK sign to assess median nerv**



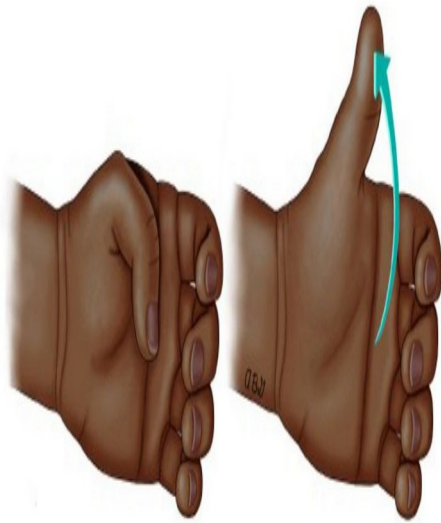
The OK sign evaluates the anterior interosseous branch of the median nerve by evaluating flexion at the distal interphalangeal (DIP) joint and testing the strength of the flexor pollicis longus and flexor digitorum profundus. The patient is asked to make the OK sign and then maintain it against resistance. Patients with anterior interosseous nerve deficits will have flattening at the DIP joint and weakness of the muscles of apposition.

### Ulnar nerve test



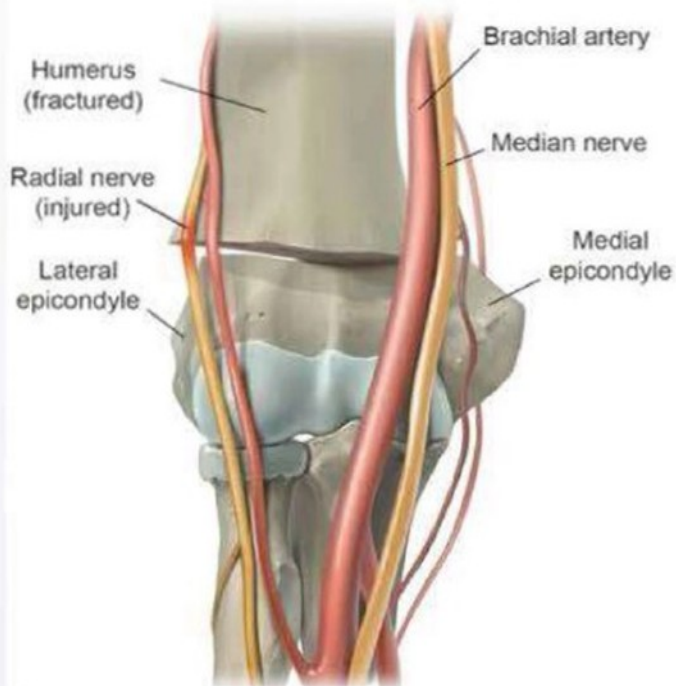
The examiner attempts to remove a piece of paper being held between the patient's fingers. This method tests the motor function of the lumbrical muscles that are innervated by the ulnar nerve.

### Thumbs up sign

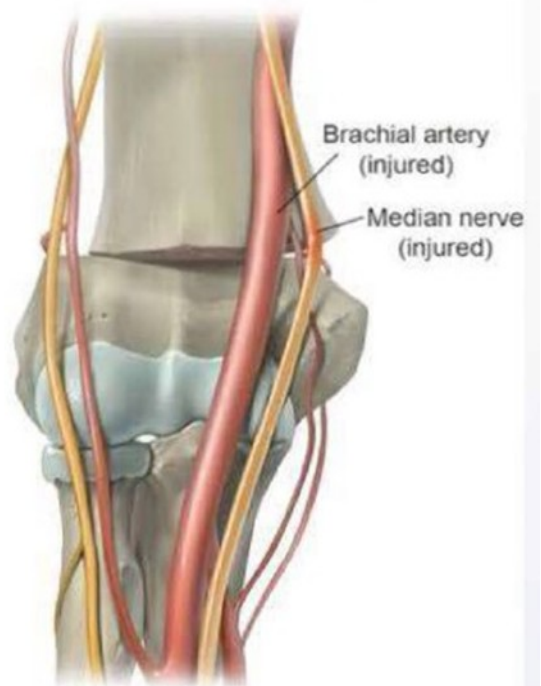


This thumb motion indicates intact radial nerve motor function.

### Anterolateral humeral fracture



### Anteromedial humeral fracture



**Case scenario 2: 6-year-old girl presents with pain in her lt. elbow for 1 hr. duration.**

**1- Take history from her**

**2- Perform physical examination of lt. elbow joint**

**3- Formulate a differential diagnosis**

**SP instruction**

|                                              |                                                                                                                                                                                                                                                                                                                                                                           |
|----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Scenario</b>                              | Zainab Ali , 6-year-old girl                                                                                                                                                                                                                                                                                                                                              |
| <b>History of present illness</b>            | Chief complaint: left elbow pain                                                                                                                                                                                                                                                                                                                                          |
| Duration                                     | 1 hr.                                                                                                                                                                                                                                                                                                                                                                     |
| Precipitating events and mechanism of injury | Pain immediately after falling off the trampoline during playing in the park falling on outstretched hand                                                                                                                                                                                                                                                                 |
| Aggravating/alleviating factors              | Constant pain                                                                                                                                                                                                                                                                                                                                                             |
| Character                                    | sharp                                                                                                                                                                                                                                                                                                                                                                     |
| Severity                                     | Sever 9/10                                                                                                                                                                                                                                                                                                                                                                |
| Course                                       | Progressive                                                                                                                                                                                                                                                                                                                                                               |
| Associated symptoms                          | Swelling and bluish discoloration                                                                                                                                                                                                                                                                                                                                         |
| <b>Current life situation</b>                | (where does patient live/work, ...):                                                                                                                                                                                                                                                                                                                                      |
| <b>Past medical history</b>                  | none                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Family history</b>                        | none                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Medications</b>                           | none. No drug allergy                                                                                                                                                                                                                                                                                                                                                     |
| <b>Physical exam</b>                         | Joint examination reveals S shaped deformity, marked swelling and ecchymosis. Marked tenderness over the medial and lateral epicondyle of the humerus. Reluctant to move her elbow.<br><b>There are weak brachial and radial artery pulses.</b><br><b>Sensory loss over the palmer index finger. Flattening of thumb and index fingers DIP joints flexion (Okay sign)</b> |

